

① Evaluate $\int_{-\pi}^{\pi} (\cos x - \sin x)^2 dx$

② Write the direction ratio of the following line
 $x = -3, \frac{y-4}{3} = \frac{z-2}{1}$

③ $\int \frac{x^2 - 3x + 1}{\sqrt{1-x^2}} dx$

④ Show that the differential equation is homogeneous and solve it.

$$2xy \frac{dy}{dx} = x^2 + 3y^2.$$

⑤ Let $A = \mathbb{R} - \{2\}$ and $B = \mathbb{R} - \{1\}$. If $f: A \rightarrow B$ is a function defined by

$$f(x) = \frac{x-1}{x-2}$$

Show that f is one-one and onto. Hence, find $f^{-1}(x)$.

⑥ Find the equation of tangent to the curve $y = \sqrt{3x-2}$ which is parallel to the line $4x - 2y + 5 = 0$.

⑦ Find the intervals in which the function f given by $f(x) = \sin x + \cos x, 0 \leq x \leq 2\pi$ is strictly increasing or strictly decreasing.

⑧ Find the value of k so that the following function is continuous at $x = 2$

$$f(x) = \begin{cases} \frac{x^3 + x^2 - 16x + 20}{(x-2)^2} & ; x \neq 2 \\ k & ; x = 2 \end{cases}$$